Fifth Implementation Iteration: Space Allocation

with One-Time and Recurring Requests

Software Design / Iteration Report

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1. Iteration Goal

The goal of this fifth implementation iteration is to extend the prototype by adding a second request type: `RecurringRequest`. In previous iterations, the system supported only one-time requests for multiple space types. In this iteration, the system is extended to process repeated booking requests for the same space across multiple time slots.

The existing decision rules remain unchanged. A recurring request is approved only if the requested space satisfies the capacity requirement and all requested occurrences satisfy the availability rule.

2. Changes from Iteration 4

Compared with the fourth iteration, the following changes were introduced:

- Added a new request type: `RecurringRequest`
- Extended the request-processing logic to support multiple occurrences
- Updated the availability checking mechanism so that all requested time slots of a recurring request are evaluated
- Preserved the same supported space types:
  - `Classroom`
  - `Laboratory`
  - `MeetingRoom`
- Preserved the same rules:
  - `AvailabilityRule`
  - `CapacityRule`

3. Scenario

A user submits either a one-time request or a recurring request for a specific space resource. A one-time request contains a single time slot, while a recurring request contains multiple time slots representing repeated occurrences of the same booking.

The system checks whether the requested space is available during the requested time slot or time slots and whether the space capacity is sufficient for the participant count.

A one-time request is approved if its single occurrence passes all rules. A recurring request is approved only if all occurrences pass all rules. Otherwise, the request is rejected.

4. Modules and Their Tasks

A. Request Management Module

- Receive one-time or recurring space requests from the user
- Create and store `OneTimeRequest` or `RecurringRequest`
- Include request details:
  - requester
  - requested space
  - requested time slot or time slots
  - participant count
- Send the request to the Rule Evaluation Module

B. Resource Management Module

- Manage all supported space types
- Provide information for `Classroom`, `Laboratory`, and `MeetingRoom`
- Provide identity, type, name, and capacity of the requested space

C. Rule Evaluation Module

- Apply `AvailabilityRule`
- Apply `CapacityRule`
- For recurring requests, evaluate all requested occurrences
- Return the final decision: Approved or Rejected

D. Allocation Management Module

- Create one allocation for a one-time request if all rules pass
- Create multiple allocations for a recurring request if all occurrences pass
- Update request status to Approved or Rejected

5. Iteration Comparison

Iteration	Request Types	Resource Types	Rules Implemented
Iteration 1	OneTimeRequest	Classroom	AvailabilityRule
Iteration 2	OneTimeRequest	Classroom	AvailabilityRule + CapacityRule
Iteration 3	OneTimeRequest	Classroom + Laboratory	AvailabilityRule + CapacityRule
Iteration 4	OneTimeRequest	Classroom + Laboratory + MeetingRoom	AvailabilityRule + CapacityRule
Iteration 5	OneTimeRequest + RecurringRequest	Classroom + Laboratory + MeetingRoom	AvailabilityRule + CapacityRule

6. Updated Class Diagram

Figure 1 shows the updated simplified class diagram for the fifth implementation iteration.

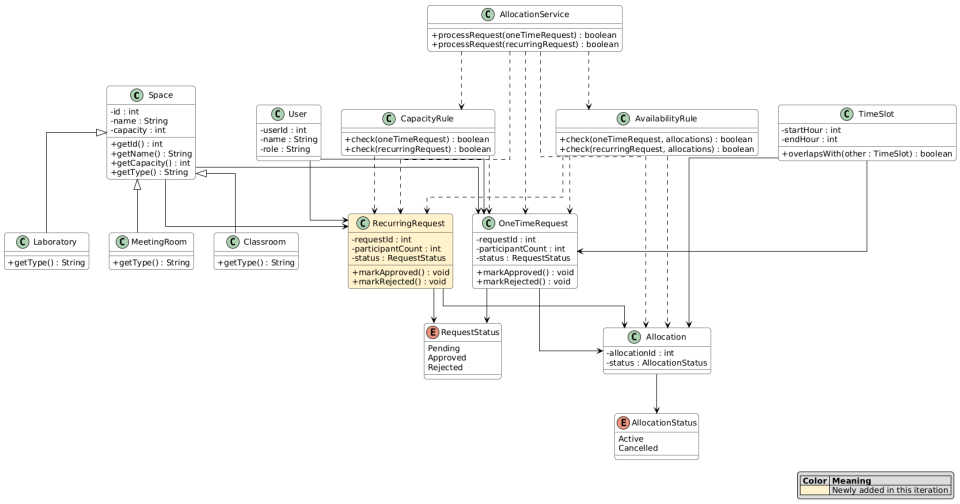

Figure 1: Updated Simplified Class Diagram

7. Updated Workflow Diagram

Figure 2 shows the workflow for the fifth iteration of the space allocation process.

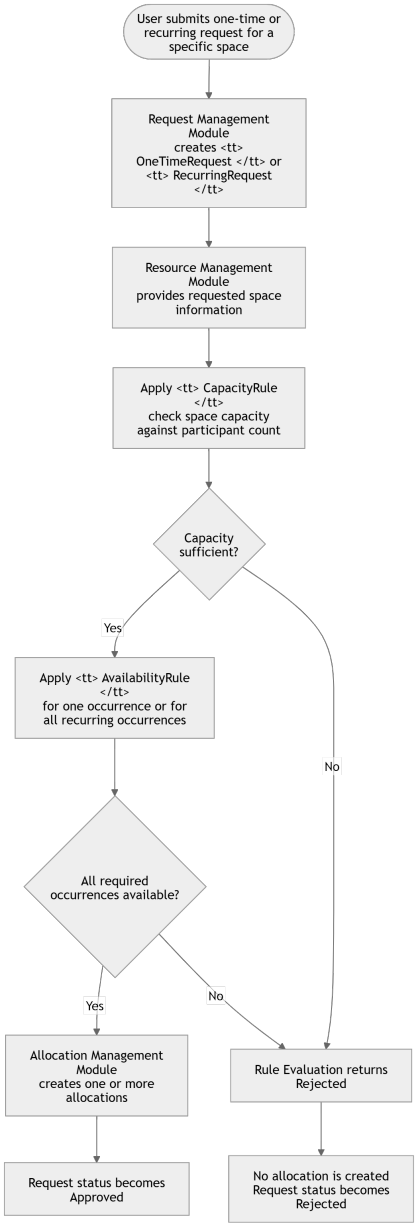

Figure 2: Updated Workflow Diagram

8. Prototype Implementation

To validate the design of this iteration, the C++ prototype was extended with a new request type, `RecurringRequest`. The allocation service was updated to support both one-time and recurring requests. For recurring requests, the service checks the capacity rule once and the availability rule across all requested occurrences.

If all occurrences pass, multiple allocations are created, one for each occurrence. If one or more occurrences fail, the recurring request is rejected.

9. Summary

This fifth implementation iteration extends the prototype by introducing a second request type while preserving the existing space types and decision rules. The system now supports both one-time and recurring allocation requests.

This iteration demonstrates controlled iterative development by extending the request model while preserving the same allocation architecture. Future iterations can continue this approach by adding more complex recurring patterns, approval rules, or scheduling constraints.